

Temporal Triangulation Cascade Protocol

TTCP v1.0.0

Tripolar Topological Crystallization

via Cascading Persistent Homology over Fiber Bundles

System class: Deterministic cascading triangulation protocol
with tripolar convergence crystallization

Objective: Detect topologically stable structures across
spatial, temporal, and meta-topological scales

Mathematical core: Persistent homology, fiber-bundle decomposition,
Wasserstein metric, spectral cascade convergence

Arithmetic: Deterministic fixed-point ($\mathbb{Q}_{32}$), zero float

Dependencies: 5D Mathematics Master, TMCP, DIC framework

Prepared for: Sebastian Klemm

Date: 11 March 2026

Protocol Identity

TTCP is a closed, deterministic protocol. Every object is defined, every step is computable, every convergence criterion is testable. The protocol produces cryptographically certified artifacts (Crystals) when and only when tripolar convergence is achieved. No hidden state, no probabilistic steps, no floating-point in identity-bearing computation.

Contents

1 Purpose and Scope	2
1.1 What This Protocol Does	2
1.2 What This Protocol Does Not Do	2
1.3 Notation	2
2 Arithmetic Foundation	2
3 The Embedding Space H_5	3
4 Persistent Homology Machinery	3
5 Level 1: Extrinsic Triangulation (5D)	4
6 Level 2: Intrinsic Fiber Triangulation (4D)	5
7 Level 3: Meta-Triangulation (Topology of Topologies)	5
8 Tripolar Convergence and Crystallization	6
8.1 The Three Poles	6
8.2 Convergence Predicate	6
9 Crystal Artifact	7
10 Evidence Chain	8
11 The Complete Protocol	9
12 Domain Adapter Interface	10
13 Configuration Schema	10
14 Computational Complexity	11
15 Spectral Interpretation (Leray-Serre Alignment)	11
16 Integration with TMCP	12
17 Invariants	12
18 Acceptance Tests	12
19 Implementation Notes	13
19.1 Recommended Libraries	13
19.2 Memory Budget	13
19.3 Parallelism	13
A Proof: Wasserstein Convergence of Cascading Levels	14
B Reference: Mapping to unified_<code>_triangulation.py</code>	14

1 Purpose and Scope

1.1 What This Protocol Does

TTCP takes a time-indexed stream of observations in a 5-dimensional embedding space H_5 and determines whether the stream contains **topologically stable structure** — not at one moment, but across three simultaneous scales:

1. The spatial topology of the full 5D point cloud (**extrinsic pole**).
2. The temporal evolution of 4D fiber topologies as the observation parameter varies (**intrinsic pole**).
3. The meta-stability of the topological evolution itself (**cyclic pole**).

When all three poles converge simultaneously, the protocol emits a **Temporal Crystal** — a hash-certified, immutable artifact recording the triple-convergence event.

1.2 What This Protocol Does Not Do

- It does not predict the future.
- It does not classify signals by content.
- It does not require training data.
- It does not use floating-point arithmetic in any identity-bearing computation.
- It does not depend on a specific domain — the embedding axes are parameterized via a Domain Adapter.

1.3 Notation

Symbol	Meaning
H_5	5-dimensional embedding space over $\mathbb{Q}_{32}$
$\mathbb{Q}_{32}$	Signed 32.32 fixed-point field (i64 backing)
$\mathcal{K}$	Simplicial complex
$\beta_k(\mathcal{K})$	k -th Betti number of $\mathcal{K}$
$\text{PD}_k(\mathcal{K})$	k -th persistence diagram of $\mathcal{K}$
W_p	p -Wasserstein distance between persistence diagrams
π	Projection map $H_5 \rightarrow B$ (base space)
F_t	Fiber $\pi^{-1}(t)$: 4D slice at parameter t
w	Windnarbe (nullcenter traversal counter)
$\mathcal{W}$	Windnarbe pair (w, χ) with rolling digest χ
$\Phi(x) = (\psi, \rho, \omega)$	Evaluation triplet: quality, stability, efficiency
δ_k	Convergence distance at step k
Θ	Crystallization threshold

2 Arithmetic Foundation

Definition 2.1 (Fixed-Point Field $\mathbb{Q}_{32}$). All numerical computation in this protocol is performed in the fixed-point field $\mathbb{Q}_{32}$: signed 64-bit integers interpreted as $q/2^{32}$, with range $[-2^{31}, 2^{31} - 2^{-32}]$ and resolution $2^{-32} \approx 2.33 \times 10^{-10}$. Division uses banker's rounding (round-half-to-even).

Arithmetic Law

No floating-point arithmetic participates in any identity-bearing computation. Two implementations on different hardware, given identical inputs and configuration, must produce bit-identical outputs. This is a hard requirement, not a goal.

Definition 2.2 (Canonical Digest). For any data structure D , its canonical digest is:

$$\text{Hash}(D) = \text{BLAKE3}(\text{CanonicalBytes}(D))$$

where `CanonicalBytes` is a deterministic, platform-independent serialization using little-endian byte order and fixed-width integer representations.

3 The Embedding Space H_5

Definition 3.1 (Embedding Space). The embedding space is:

$$H_5 = \mathbb{Q}_{32}^5$$

A point $e = (a_1, a_2, a_3, a_4, a_5) \in H_5$ represents the essential characteristics of an observation along five independent measurement axes.

Definition 3.2 (Axis Semantics). The five axes are domain-parameterized via a **Domain Adapter** (Section 12). Each axis a_i is computed by a deterministic function $f_i : \mathcal{O} \times W_i \rightarrow \mathbb{Q}_{32}$ where $\mathcal{O}$ is the observation type and W_i is a weight vector. Requirements:

1. **Determinism**: identical inputs produce identical outputs.
2. **Self-containment**: f_i depends only on the observation, not on external state.
3. **Independence**: the five axes capture qualitatively distinct aspects.

Definition 3.3 (Resonance Kernel). The factorized Gaussian RBF kernel on H_5 is:

$$K(e_x, e_y; \sigma) = \prod_{i=1}^5 \exp\left(-\frac{(x_i - y_i)^2}{2\sigma_i^2}\right)$$

with bandwidth parameters $\sigma = (\sigma_1, \dots, \sigma_5) \in \mathbb{Q}_{32}^5$, $\sigma_i > 0$. The kernel satisfies $K \in [0, 1]$ with $K(e, e) = 1$.

Definition 3.4 (Kernel Distance).

$$\delta(e_x, e_y; \sigma) = \sqrt{2 - 2K(e_x, e_y; \sigma)}$$

This is a pseudometric on H_5 : $\delta(e, e) = 0$, $\delta(e_x, e_y) = \delta(e_y, e_x)$, and the triangle inequality holds.

4 Persistent Homology Machinery

Definition 4.1 (Simplicial Complex from Point Cloud). Given a finite point set $S \subset H_5$ with $|S| = n$, the **Vietoris-Rips complex** at scale ε is:

$$\text{VR}_\varepsilon(S) = \{\sigma \subseteq S \mid \delta(x, y) \leq \varepsilon \ \forall x, y \in \sigma\}$$

Alternatively, the **alpha complex** (Delaunay-based) may be used when n is large, since it is homotopy-equivalent and computationally cheaper.

Definition 4.2 (Filtration). A filtration is a nested sequence of simplicial complexes:

$$\emptyset = \mathcal{K}_0 \subseteq \mathcal{K}_1 \subseteq \dots \subseteq \mathcal{K}_m = \mathcal{K}$$

indexed by increasing scale parameter $\varepsilon_0 < \varepsilon_1 < \dots < \varepsilon_m$.

Definition 4.3 (Persistence Diagram). For a filtration $\{\mathcal{K}_i\}$ and homology dimension k , the k -th persistence diagram PD_k is the multiset of pairs (b_j, d_j) where b_j is the birth scale and d_j is the death scale of the j -th k -homology class. Points far from the diagonal $b = d$ represent persistent (stable) topological features; points near the diagonal represent noise.

Definition 4.4 (p -Wasserstein Distance on Persistence Diagrams). For two persistence diagrams D_1, D_2 , the p -Wasserstein distance is:

$$W_p(D_1, D_2) = \left(\inf_{\gamma} \sum_{x \in D_1} \|x - \gamma(x)\|_{\infty}^p \right)^{1/p}$$

where the infimum is over all bijections $\gamma : D_1 \rightarrow D_2$ (augmented by diagonal points). We use $p = 2$ throughout.

Proposition 4.5 (Stability of Persistent Homology). For two point clouds $S_1, S_2 \subset H_5$, if the Hausdorff distance $d_H(S_1, S_2) \leq \eta$, then:

$$W_{\infty}(\text{PD}_k(S_1), \text{PD}_k(S_2)) \leq \eta$$

This is the Persistence Stability Theorem (Cohen-Steiner, Edelsbrunner, Harer, 2007). It guarantees that small perturbations in input produce small perturbations in topological output.

Definition 4.6 (Betti Signature). The Betti signature of a complex $\mathcal{K}$ at scale ε is the vector:

$$\beta(\mathcal{K}, \varepsilon) = (\beta_0, \beta_1, \beta_2, \beta_3, \beta_4) \in \mathbb{Z}_{\geq 0}^5$$

where $\beta_k = \text{rank } H_k(\mathcal{K}_{\varepsilon}; \mathbb{Z})$ and the maximum dimension is 4 (since we operate in H_5 , simplices have at most 5 vertices $\Rightarrow$ dimension ≤ 4).

5 Level 1: Extrinsic Triangulation (5D)

The first pole operates on the **full** 5D point cloud accumulated over a time window.

Definition 5.1 (Observation Window). An observation window of length N is an ordered sequence:

$$\mathcal{S}_N = \{e(t_1), e(t_2), \dots, e(t_N)\} \subset H_5$$

where $t_1 < t_2 < \dots < t_N$ and each $e(t_i)$ is the 5D embedding of the observation at time t_i .

Algorithm L1: Extrinsic Triangulation

Input: Observation window $\mathcal{S}_N$, scale parameters $\varepsilon_{\min}, \varepsilon_{\max}, n_{\varepsilon}$.

Output: Persistence diagrams $\text{PD}_k^{(5)}$ for $k = 0, \dots, 4$; Betti signature $\beta^{(5)}$; digest h_1 .

1. Construct filtration: $\{\text{VR}_{\varepsilon_j}(\mathcal{S}_N)\}_{j=0}^{n_{\varepsilon}}$ with $\varepsilon_j = \varepsilon_{\min} + j \cdot (\varepsilon_{\max} - \varepsilon_{\min}) / n_{\varepsilon}$.
2. Compute persistent homology via boundary matrix reduction (standard algorithm).
3. Extract $\text{PD}_k^{(5)}$ for each dimension k .
4. Compute Betti signature $\beta^{(5)}$ at a reference scale $\varepsilon^* = (\varepsilon_{\min} + \varepsilon_{\max}) / 2$.
5. Compute digest: $h_1 = \text{Hash}(\text{PD}_0^{(5)}, \dots, \text{PD}_4^{(5)}, \beta^{(5)})$.
6. **Return** $(\text{PD}^{(5)}, \beta^{(5)}, h_1)$.

Definition 5.2 (L1 Convergence Metric). Given two consecutive observation windows $\mathcal{S}_N^{(w)}$ and $\mathcal{S}_N^{(w+1)}$ (where w is the window counter), the L1 convergence distance is:

$$\delta_1(w) = \sum_{k=0}^4 \alpha_k \cdot W_2\left(\text{PD}_k^{(5)}(w), \text{PD}_k^{(5)}(w+1)\right)$$

where $\alpha_k \geq 0$ are configured dimension weights with $\sum_k \alpha_k = 1$.

6 Level 2: Intrinsic Fiber Triangulation (4D)

The second pole decomposes the 5D cloud into 4D fibers over a base parameter, then tracks how fiber topology evolves.

Definition 6.1 (Projection and Fiber Bundle). Fix a projection axis $\pi_a : H_5 \rightarrow \mathbb{Q}_{32}$ extracting the a -th coordinate ($a \in \{1, \dots, 5\}$, configurable; default: $a = 5$, the entropy/temporal axis). The projection partitions the observation window into fibers:

$$F_j = \{e \in \mathcal{S}_N \mid \pi_a(e) \in [b_j, b_{j+1})\}, \quad j = 0, \dots, M_f - 1$$

where $[b_0, b_{M_f})$ covers the range of π_a over $\mathcal{S}_N$ and M_f is the number of fiber slices.

Remark 6.2. The fiber F_j is a subset of a 4-dimensional affine subspace of H_5 (the hyperplane $a_a \approx \bar{b}_j$). Its triangulation is intrinsically 4-dimensional.

Algorithm L2: Fiber Triangulation Sequence

Input: Observation window $\mathcal{S}_N$, projection axis a , fiber count M_f , scale parameters.

Output: Fiber persistence sequence $\{\text{PD}_k^{(4)}(j)\}_{j,k}$; Betti trajectory $\{\beta^{(4)}(j)\}_j$; digest h_2 .

1. Compute projection range: $[v_{\min}, v_{\max}] = [\min_{e \in \mathcal{S}_N} \pi_a(e), \max_{e \in \mathcal{S}_N} \pi_a(e)]$.
2. Partition into M_f equi-width bins: $b_j = v_{\min} + j \cdot (v_{\max} - v_{\min}) / M_f$.
3. For each fiber $j = 0, \dots, M_f - 1$:
 - (a) Extract fiber points: $F_j = \{e \in \mathcal{S}_N \mid \pi_a(e) \in [b_j, b_{j+1})\}$.
 - (b) Project to 4D: remove coordinate a from each point.
 - (c) If $|F_j| < 6$: skip (insufficient for meaningful topology). Mark $\beta^{(4)}(j) = \mathbf{0}$.
 - (d) Else: compute VR filtration on F_j in $\mathbb{Q}_{32}^4$.
 - (e) Extract $\text{PD}_k^{(4)}(j)$ for $k = 0, \dots, 3$ and Betti signature $\beta^{(4)}(j)$.
4. Assemble Betti trajectory: $\mathcal{B} = (\beta^{(4)}(0), \beta^{(4)}(1), \dots, \beta^{(4)}(M_f - 1))$.
5. Compute digest: $h_2 = \text{Hash}(\mathcal{B}, \{\text{PD}_k^{(4)}(j)\}_{j,k})$.
6. **Return** $(\{\text{PD}_k^{(4)}(j)\}_{j,k}, \mathcal{B}, h_2)$.

Definition 6.3 (L2 Convergence Metric). The L2 convergence distance measures how much fiber topology changes across consecutive fibers:

$$\delta_2(w) = \frac{1}{M_f - 1} \sum_{j=0}^{M_f-2} \sum_{k=0}^3 \alpha_k \cdot W_2\left(\text{PD}_k^{(4)}(j), \text{PD}_k^{(4)}(j+1)\right)$$

A small δ_2 means fiber topology is approximately constant across the base parameter — the bundle is “flat.”

7 Level 3: Meta-Triangulation (Topology of Topologies)

The third pole triangulates **the space of persistence diagrams itself**, treating each fiber’s topological summary as a point in a derived metric space.

Definition 7.1 (Persistence Summary Vector). For fiber j , define its persistence summary vector:

$$\mathbf{p}(j) = \left(\beta_0^{(4)}(j), \beta_1^{(4)}(j), \beta_2^{(4)}(j), \beta_3^{(4)}(j), \text{Pers}(j) \right) \in \mathbb{Q}_{32}^5$$

where $\text{Pers}(j) = \sum_{(b,d) \in \text{PD}_*^{(4)}(j)} (d-b)$ is the total persistence of fiber j (sum over all dimensions).

Remark 7.2. Each $\mathbf{p}(j)$ is itself a point in $\mathbb{Q}_{32}^5$. The collection $\{\mathbf{p}(j)\}_{j=0}^{M_f-1}$ forms a **point cloud in the derived 5D space of topological summaries**. This is the key insight: we can triangulate this derived cloud with the *same* persistent homology machinery.

Algorithm L3: Meta-Triangulation

Input: Persistence summary vectors $\{\mathbf{p}(j)\}_{j=0}^{M_f-1}$ from L2; scale parameters.

Output: Meta-persistence diagrams $\text{PD}_k^{(\text{meta})}$; meta-Betti signature $\beta^{(\text{meta})}$; digest h_3 .

1. Treat $\mathcal{P} = \{\mathbf{p}(0), \dots, \mathbf{p}(M_f - 1)\}$ as a point cloud in $\mathbb{Q}_{32}^5$.
2. Construct VR filtration on $\mathcal{P}$ using the kernel distance $\delta(\cdot, \cdot; \sigma_{\text{meta}})$ with a dedicated meta-bandwidth σ_{meta} .
3. Compute persistent homology: extract $\text{PD}_k^{(\text{meta})}$ for $k = 0, \dots, 4$.
4. Compute $\beta^{(\text{meta})}$ at reference scale.
5. Compute digest: $h_3 = \text{Hash}(\text{PD}^{(\text{meta})}, \beta^{(\text{meta})})$.
6. **Return** $(\text{PD}^{(\text{meta})}, \beta^{(\text{meta})}, h_3)$.

Definition 7.3 (L3 Convergence Metric). Given meta-triangulations at windnarbe steps w and $w + 1$:

$$\delta_3(w) = \sum_{k=0}^4 \alpha_k \cdot W_2\left(\text{PD}_k^{(\text{meta})}(w), \text{PD}_k^{(\text{meta})}(w + 1)\right)$$

Proposition 7.4 (Recursive Applicability). Since $\mathbf{p}(j) \in \mathbb{Q}_{32}^5$ and the triangulation machinery operates on any finite point cloud in $\mathbb{Q}_{32}^5$, Levels 1–3 use **the same algorithm** (VR filtration $\rightarrow$ persistent homology $\rightarrow$ Betti signature $\rightarrow$ digest). The only difference is the input: Level 1 receives raw observations, Level 2 receives fiber slices, Level 3 receives topological summaries.

8 Tripolar Convergence and Crystallization

8.1 The Three Poles

Pole	Name	What it measures	TMCP carrier analogy
P1	Extrinsic	Is the 5D spatial topology stable?	t_1 (irreversible commit)
P2	Intrinsic	Is fiber topology constant across the base?	t_2 (exploratory evolution)
P3	Cyclic	Is the pattern of topological change itself stable?	ϕ (phase drift)

8.2 Convergence Predicate

Definition 8.1 (Condensation Window). A condensation window of size s at step w is the sequence $(\delta_p(w - s + 1), \dots, \delta_p(w))$ for pole $p \in \{1, 2, 3\}$.

Definition 8.2 (Pole Convergence). Pole p is **converged** at step w with window s if:

$$\text{converged}_p(w, s) \iff \forall j \in [w - s + 2, w] : \delta_p(j) \leq \delta_p(j - 1)$$

That is, the convergence distance has been monotonically non-increasing over the entire window.

Definition 8.3 (Tripolar Crystallization Condition). The protocol declares **crystallization** at step w if and only if all three conditions hold simultaneously:

$$\text{Crystal}(w) \iff \text{converged}_1(w, s) \wedge \text{converged}_2(w, s) \wedge \text{converged}_3(w, s) \wedge \max_p \delta_p(w) \leq \Theta \quad (1)$$

where $\Theta > 0$ is the crystallization threshold and $s \geq s_{\min}$ is the minimum window size.

Theorem 8.4 (Crystallization Convergence). Let $\{e(t_i)\}_{i=1}^\infty$ be an infinite observation stream from a process whose 5D embedding trajectory converges to an attractor $A \subset H_5$ (i.e., $\lim_{i \rightarrow \infty} \text{dist}(e(t_i), A) = 0$). Then:

1. By the Persistence Stability Theorem (Prop. 4.5), $\delta_1(w) \rightarrow 0$ as $w \rightarrow \infty$.
2. By the same theorem applied to fibers, $\delta_2(w) \rightarrow 0$.
3. By the same theorem applied to the derived cloud, $\delta_3(w) \rightarrow 0$.
4. Therefore, for any $\Theta > 0$ and $s_{\min} \geq 1$, there exists w_0 such that $\text{Crystal}(w_0) = \text{true}$.

Proof. Since $e(t_i) \rightarrow A$, the Hausdorff distance between consecutive windows satisfies $d_H(\mathcal{S}_N^{(w)}, \mathcal{S}_N^{(w+1)}) \rightarrow 0$. By the Persistence Stability Theorem, $W_\infty(\text{PD}_k^{(5)}(w), \text{PD}_k^{(5)}(w+1)) \leq d_H(\mathcal{S}_N^{(w)}, \mathcal{S}_N^{(w+1)}) \rightarrow 0$, hence $\delta_1(w) \rightarrow 0$.

For L2: each fiber F_j converges to $F_j \cap A$, so by the same argument $\text{PD}_k^{(4)}(j; w) \rightarrow \text{PD}_k^{(4)}(j; \infty)$, and $\delta_2(w) \rightarrow 0$.

For L3: since each $\mathbf{p}(j; w) \rightarrow \mathbf{p}(j; \infty)$, the derived cloud converges, and $\delta_3(w) \rightarrow 0$.

Since all three sequences converge to 0, they are eventually monotonically non-increasing and eventually below Θ . $\square$

Remark 8.5. The theorem says: if the process stabilizes, the protocol *will* detect it. If the process does not stabilize, the protocol correctly never crystallizes. There are no false positives (by construction — all three poles must agree) and no missed true positives (by the convergence theorem).

9 Crystal Artifact

Definition 9.1 (Temporal Crystal). A Temporal Crystal emitted at windnarbe step w_0 is the tuple:

$$\mathcal{C} = (w_0, \beta^{(5)}, \mathcal{B}, \beta^{(\text{meta})}, \delta_1(w_0), \delta_2(w_0), \delta_3(w_0), h_1, h_2, h_3, h_{\mathcal{C}})$$

where:

- w_0 : windnarbe at crystallization.
- $\beta^{(5)}, \mathcal{B}, \beta^{(\text{meta})}$: the three Betti-level summaries.
- $\delta_1, \delta_2, \delta_3$: final convergence distances.
- h_1, h_2, h_3 : level digests.
- $h_{\mathcal{C}} = \text{Hash}(w_0, h_1, h_2, h_3, \delta_1, \delta_2, \delta_3)$: crystal digest.

Crystal Integrity Law

A Crystal is valid if and only if $h_{\mathcal{C}}$ verifies against the contained data. Any modification to any field invalidates the digest. The Crystal is immutable after emission.

10 Evidence Chain

Definition 10.1 (TTCP Event). A TTCP event is a typed record:

$$\varepsilon = (\text{tag}, \text{payload}, w, h_{\text{prev}})$$

where $\text{tag} \in \{\text{L1_Complete}, \text{L2_Complete}, \text{L3_Complete}, \text{Crystal}, \text{NoConverge}, \text{Fault}\}$, w is the current windnarbe, and h_{prev} is the previous evidence digest.

Definition 10.2 (Hash-Chained Evidence).

$$h_0 = \text{Hash}(\text{"TTCP_GENESIS"}), \quad h_i = \text{Hash}(\text{tag}_i \parallel \text{payload}_i \parallel w_i \parallel h_{i-1})$$

Theorem 10.3 (No Phantom Crystallizations). Under deterministic replay and canonical encoding, any asserted Crystal must have a corresponding hash-chained evidence sequence containing L1, L2, L3 completion events and a Crystal event whose windnarbe matches. Absence of any link in the chain is proof of non-occurrence.

11 The Complete Protocol

Algorithm 1 TTCP Core Loop

Require: Observation stream $\{e(t)\}$, config C , genesis state
Ensure: Evidence chain, Crystals (if convergence achieved)

- 1: Initialize $w \leftarrow 0$, $h_{\text{prev}} \leftarrow \text{Hash}(\text{"TTCP_GENESIS"})$
- 2: Initialize history buffers $\Delta_1[], \Delta_2[], \Delta_3[]$
- 3: **while** observation stream active **do**
- 4: Ingest N new observations into window $\mathcal{S}_N$
- 5: $w \leftarrow w + 1$
- 6: $\triangleright$ — **Level 1: Extrinsic (5D)** —
- 7: $(\text{PD}^{(5)}, \beta^{(5)}, h_1) \leftarrow \text{L1_TRIANGULATE}(\mathcal{S}_N, C)$
- 8: **if** $w > 1$ **then**
- 9: $\delta_1(w) \leftarrow \text{WASSERSTEINDIST}(\text{PD}_{\text{prev}}^{(5)}, \text{PD}^{(5)})$
- 10: Append $\delta_1(w)$ to Δ_1
- 11: **end if**
- 12: Emit event (**L1_Complete**, h_1, w, h_{prev}); update h_{prev}
- 13: $\triangleright$ — **Level 2: Intrinsic (4D fibers)** —
- 14: $(\{\text{PD}_j^{(4)}\}, \mathcal{B}, h_2) \leftarrow \text{L2_FIBERTRIANGULATE}(\mathcal{S}_N, C)$
- 15: $\delta_2(w) \leftarrow \text{MEANFIBERDIST}(\{\text{PD}_j^{(4)}\})$
- 16: Append $\delta_2(w)$ to Δ_2
- 17: Emit event (**L2_Complete**, h_2, w, h_{prev}); update h_{prev}
- 18: $\triangleright$ — **Level 3: Meta-triangulation** —
- 19: $\{\mathbf{p}(j)\} \leftarrow \text{PERSISTENCESUMMARYVECTORS}(\{\text{PD}_j^{(4)}\}, \mathcal{B})$
- 20: $(\text{PD}^{(\text{meta})}, \beta^{(\text{meta})}, h_3) \leftarrow \text{L3_METATRIANGULATE}(\{\mathbf{p}(j)\}, C)$
- 21: **if** $w > 1$ **then**
- 22: $\delta_3(w) \leftarrow \text{WASSERSTEINDIST}(\text{PD}_{\text{prev}}^{(\text{meta})}, \text{PD}^{(\text{meta})})$
- 23: Append $\delta_3(w)$ to Δ_3
- 24: **end if**
- 25: Emit event (**L3_Complete**, h_3, w, h_{prev}); update h_{prev}
- 26: $\triangleright$ — **Tripolar Convergence Check** —
- 27: **if** $|\Delta_1| \geq s_{\min}$ **and** $|\Delta_2| \geq s_{\min}$ **and** $|\Delta_3| \geq s_{\min}$ **then**
- 28: **if** $\text{CONVERGED}(\Delta_1, s) \wedge \text{CONVERGED}(\Delta_2, s) \wedge \text{CONVERGED}(\Delta_3, s)$ **then**
- 29: **if** $\max(\delta_1(w), \delta_2(w), \delta_3(w)) \leq \Theta$ **then**
- 30: $\mathcal{C} \leftarrow \text{EMITCRYSTAL}(w, \beta^{(5)}, \mathcal{B}, \beta^{(\text{meta})}, \delta_1, \delta_2, \delta_3, h_1, h_2, h_3)$
- 31: Emit event (**Crystal**, $h_{\mathcal{C}}, w, h_{\text{prev}}$); update h_{prev}
- 32: **end if**
- 33: **end if**
- 34: **end if**
- 35: Store current PDs as “prev” for next iteration
- 36: **end while**

Protocol Law

The core loop is the single entry point. All three levels execute sequentially per iteration. No level may be skipped. No crystallization may occur without all three levels completing and all three convergence predicates being satisfied.

12 Domain Adapter Interface

Listing 1: Domain Adapter trait (Rust)

```
pub trait DomainAdapter: Send + Sync {
    type Observation;

    /// Embed a raw observation into H5.
    fn embed(&self, obs: &Self::Observation) -> [Q32; 5];

    /// Axis labels for diagnostics.
    fn axis_names(&self) -> [&'static str; 5];

    /// Default kernel bandwidths.
    fn default_sigma(&self) -> [Q32; 5];
}
```

The Domain Adapter is the **only** domain-specific component. Everything else is universal. Examples:

- **Trading domain:** a_1 = price coherence, a_2 = volume stability, a_3 = spread entropy, a_4 = cross-venue consistency, a_5 = temporal regularity.
- **API traffic:** a_1 = structural coupling, a_2 = functional density, a_3 = topological complexity, a_4 = symmetry, a_5 = entropy.

13 Configuration Schema

Listing 2: TTCP configuration

```
ttcp:
  # Observation
  window_size: 200          # N: points per observation window
  window_stride: 50         # overlap between consecutive windows

  # Level 1 (5D)
  l1_epsilon_min: 100       # Q32-scaled minimum VR scale
  l1_epsilon_max: 10000     # Q32-scaled maximum VR scale
  l1_epsilon_steps: 50      # filtration resolution
  l1_dimension_weights: [0.3, 0.3, 0.2, 0.1, 0.1] # alpha_k

  # Level 2 (4D fibers)
  l2_projection_axis: 5     # which axis to fiber over (1-5)
  l2_fiber_count: 16        # M_f: number of fiber slices
  l2_min_fiber_points: 6    # skip fibers with fewer points

  # Level 3 (meta)
  l3_meta_sigma: [1000, 1000, 1000, 1000, 5000] # meta-bandwidth

  # Convergence
  condensation_window: 5    # s_min: minimum window for convergence
  crystallization_threshold: 50 # Theta (Q32-scaled)

  # Evidence
  hash_algorithm: blake3
  genesis_string: "TTCP_GENESIS"
```

```
# Arithmetic
q32_epsilon: 1          # smallest representable positive value
```

Configuration Law

Every numeric constant in the protocol must appear in the configuration. No magic constants in code. Configuration is hash-bound in the evidence chain via $\text{CfgHash} = \text{Hash}(\text{config})$.

14 Computational Complexity

Component	Complexity	Notes
VR filtration	$O(n^{d+1})$ where $d = \max$ simplex dim	For $n = 200$, $d = 4$: $\sim 3.2 \times 10^{11}$ worst case. Use alpha-complex for reduction.
PH computation	$O(m^3)$ where $m = \text{number of simplices}$	Standard boundary-matrix reduction.
W_2 distance	$O(p^3)$ where $p = \text{diagram size}$	Hungarian algorithm or auction.
L1 per step	$O(N^5)$ worst case	Dominated by VR construction.
L2 per step	$O(M_f \cdot (N/M_f)^4)$	Each fiber has $\sim N/M_f$ points.
L3 per step	$O(M_f^5)$	M_f points in 5D. Typically $M_f \leq 32$: trivial.

Remark 14.1. For real-time use with $N = 200$: L1 is the bottleneck. Use alpha-complexes (Delaunay-based, $O(n^{\lceil d/2 \rceil})$ in dimension d) or approximate persistent homology (e.g., SimBa, Ripser). L2 and L3 are fast because fiber sizes and M_f are small.

15 Spectral Interpretation (Leray-Serre Alignment)

The three-level cascade has a precise algebraic-topological interpretation as the first pages of a Leray-Serre spectral sequence.

Proposition 15.1 (Spectral Sequence Alignment). Let $E \xrightarrow{\pi} B$ be the fiber bundle with total space $E = \mathcal{S}_N \subset H_5$, base $B = \pi_a(\mathcal{S}_N) \subset \mathbb{Q}_{32}$, and fiber $F_b = \pi^{-1}(b)$. The Leray-Serre spectral sequence satisfies:

$$E_2^{p,q} = H_p(B; \mathcal{H}_q) \implies H_{p+q}(E)$$

where $\mathcal{H}_q$ is the local coefficient system $b \mapsto H_q(F_b)$.

The TTCP levels correspond to:

- **Level 1** computes $H_*(E)$ directly (the abutment).
- **Level 2** computes $H_q(F_b)$ for each b (the fiber homology, forming $\mathcal{H}_q$).
- **Level 3** computes an approximation to $H_p(B; \mathcal{H}_q)$ (the E_2 page).

Convergence of all three levels simultaneously is equivalent to the spectral sequence degenerating at E_2 — the simplest possible topological structure, where no higher differentials contribute.

Remark 15.2. When the spectral sequence degenerates at E_2 , the topology of the total space is completely determined by the topology of the base and fibers. This is the formal meaning of “the process is topologically transparent” — there are no hidden interactions between scales.

16 Integration with TMCP

TTCP is designed to be embedded as a subsystem within a TMCP-governed architecture:

TMCP concept	TTCP instantiation	Coupling
Observation $\rightarrow$ Multiplex	L1 + L2 triangulation	Observation window feeds L1; fiber decomposition is WT multiplexing
Kairos gate $G(x)$	Tripolar convergence check	Gate opens when $\text{Crystal}(w) = \text{true}$
Press P	L3 meta-contraction	Derived cloud contraction is a press
Crystallization $K \triangleleft NC$	Crystal emission	Hash-certified, NC-factored
Windnarbe $\mathcal{W}$	Step counter w	Incremented per cascade iteration
Evidence monoid Evaluation triplet (ψ, ρ, ω)	TTCP event chain Derived from $(\delta_1, \delta_2, \delta_3)$	Same hash-chaining structure $\psi = 1 - \delta_1/\Theta$, $\rho = 1 - \delta_2/\Theta$, $\omega = 1 - \delta_3/\Theta$

17 Invariants

TTCP Invariant Register

- INV-T1 No Crystal without all three levels completing and all three convergence predicates satisfied.
- INV-T2 No floating-point in any identity-bearing computation.
- INV-T3 Evidence chain hash-consistency: any gap or mutation is detectable.
- INV-T4 Deterministic replay: identical inputs + config = identical evidence chain + identical Crystals.
- INV-T5 Configuration immutability: CfgHash is fixed at genesis; any change requires new genesis.
- INV-T6 Fiber completeness: all M_f fibers are evaluated (skipped fibers marked, not omitted).
- INV-T7 No Crystal may be emitted with $\max(\delta_1, \delta_2, \delta_3) > \Theta$.

18 Acceptance Tests

1. **Determinism:** Two runs with identical input, config, and genesis produce bit-identical evidence chains and Crystals.
2. **Convergent process:** A synthetic signal converging to a fixed attractor must trigger crystallization within bounded steps.
3. **Non-convergent process:** A synthetic chaotic signal must never trigger crystallization (run 10,000 steps, verify zero Crystals).

4. **Fiber consistency:** For a signal with known fiber structure (e.g., product topology $S^1 \times D^4$), L2 Betti numbers must match theoretical prediction.
5. **Evidence integrity:** Mutating any event in the chain invalidates all subsequent digests.
6. **Crystal validity:** Every emitted Crystal verifies against its digest.
7. **No phantom crystals:** Removing an L2 completion event from the chain makes the Crystal event unverifiable.
8. **Wasserstein stability:** Perturbing input by η in Hausdorff distance produces $|\delta_p - \delta'_p| \leq C \cdot \eta$ for a known constant C .

19 Implementation Notes

19.1 Recommended Libraries

- **Persistent homology:** Ripser (C++/Rust bindings), GUDHI (C++/Python), or Eirene (Julia).
- **Wasserstein distance:** Hera library (C++, exact W_2), or GUDHI's `wasserstein_distance`.
- **Delaunay/Alpha complex:** CGAL (C++) or `scipy.spatial.Delaunay` (Python prototype).
- **Hashing:** BLAKE3 (Rust crate `blake3`).
- **Fixed-point:** Custom Q32 type or the `fixed` Rust crate.

19.2 Memory Budget

Per iteration: $O(N \cdot 5)$ for the point cloud + $O(m)$ for the simplicial complex + $O(p)$ for persistence diagrams. For $N = 200$: ~ 10 MB peak with alpha complexes.

19.3 Parallelism

L2 fiber triangulations are independent and can run in parallel (M_f threads). L1 and L3 are sequential but typically faster (L1 is one large complex; L3 is tiny).

A Proof: Wasserstein Convergence of Cascading Levels

Theorem A.1 (Cascade Convergence Rate). If the observation stream satisfies $d_H(\mathcal{S}_N^{(w)}, \mathcal{S}_N^{(w+1)}) \leq C \cdot r^w$ for some $C > 0$ and $r \in (0, 1)$ (geometric convergence to attractor), then:

$$\delta_1(w) \leq C_1 \cdot r^w \quad (2)$$

$$\delta_2(w) \leq C_2 \cdot r^w \quad (3)$$

$$\delta_3(w) \leq C_3 \cdot r^{2w} \quad (4)$$

where C_1, C_2 depend on the persistence stability constant and C_3 depends on the Lipschitz constant of the persistence summary map.

Proof. For L1: directly from Persistence Stability (Prop. 4.5), $\delta_1(w) \leq \sum_k \alpha_k \cdot d_H(\mathcal{S}_N^{(w)}, \mathcal{S}_N^{(w+1)}) \leq C \cdot r^w$.

For L2: each fiber satisfies $d_H(F_j^{(w)}, F_j^{(w+1)}) \leq d_H(\mathcal{S}_N^{(w)}, \mathcal{S}_N^{(w+1)})$, so the same bound applies per fiber, and averaging preserves the rate.

For L3: the persistence summary map $\mathbf{p}(j) = (\beta_0(j), \dots, \text{Pers}(j))$ is Lipschitz in the Wasserstein distance with constant L (since Betti numbers at fixed scale are locally constant and total persistence is W_1 -Lipschitz). Therefore $\|\mathbf{p}(j; w) - \mathbf{p}(j; w+1)\| \leq L \cdot C_2 \cdot r^w$. The derived cloud at step w and $w+1$ differs by at most $L \cdot C_2 \cdot r^w$ in Hausdorff distance. Applying stability again: $\delta_3(w) \leq L^2 \cdot C_2 \cdot r^w \cdot C_2 \cdot r^w = C_3 \cdot r^{2w}$.

The quadratic convergence of L3 is the “leverage effect”: cascading persistent homology squares the convergence rate. $\square$

Corollary A.2 (Super-Linear Convergence of the Tripol). The maximum convergence distance $\delta_{\max}(w) = \max(\delta_1, \delta_2, \delta_3)$ satisfies:

$$\delta_{\max}(w) \leq \max(C_1, C_2) \cdot r^w$$

while the product satisfies:

$$\delta_1(w) \cdot \delta_2(w) \cdot \delta_3(w) \leq C_1 C_2 C_3 \cdot r^{4w}$$

The multiplicative convergence of the tripol is quartic in the base convergence rate.

B Reference: Mapping to unified `_triangulation.py`

The existing Python implementation covers a subset of TTCP:

TTCP Component	Python Status	What to add
H_5 embedding	✓ (State5D)	Convert to Q32
Delaunay triangulation	✓ (scipy)	Already functional
Persistent homology	Partial (gudhi)	Complete VR filtration
Resonance kernel	✓ (edge weighting)	Already functional
Fiber decomposition (L2)	Missing	Add projection + slicing
Meta-triangulation (L3)	Missing	Add derived-cloud PH
Wasserstein distance	Missing	Add gudhi.wasserstein
Convergence tracking	Missing	Add $\delta_p(w)$ history

Tripolar crystallization	Missing	Add 3-pole check
Evidence chain	Missing	Add hash chaining
Crystal emission	Missing	Add artifact struct

End of specification TTCP v1.0.0